

The Scanner Class in Java

The `Scanner` class in Java, part of the `java.util` package, is used to read input from various sources like the keyboard (standard input), files, or strings. It provides methods to parse primitive data types and strings using regular expressions.

Features of Scanner Class

- Ease of Input Handling:**
 - Simplifies the process of taking input from the user or files.
 - Supports Multiple Data Types:**
 - Can read and parse integers, doubles, strings, booleans, etc.
 - Flexible Input Sources:**
 - Can take input from the keyboard (`System.in`), files, or strings.
 - Line and Token-Based Input:**
 - Reads input as tokens or lines, depending on the method used.
-

Importing the Scanner Class

To use the `Scanner` class, import it from the `java.util` package:

```
import java.util.Scanner;
```

Creating a Scanner Object

A `Scanner` object is created by associating it with an input source. The most common use is reading input from the keyboard:

```
Scanner scanner = new Scanner(System.in);
```

Methods of the Scanner Class

Method	Description	Example
<code>nextInt()</code>	Reads the next token as an integer.	<code>int number = scanner.nextInt();</code>
<code>nextDouble()</code>	Reads the next token as a double.	<code>double value = scanner.nextDouble();</code>
<code>nextLine()</code>	Reads the entire line as a string.	<code>String line = scanner.nextLine();</code>
<code>next()</code>	Reads the next token (word).	<code>String word = scanner.next();</code>
<code>nextBoolean()</code>	Reads the next token as a boolean.	<code>boolean flag = scanner.nextBoolean();</code>
<code>hasNextInt()</code>	Checks if the next token can be parsed as an integer.	<code>if(scanner.hasNextInt())</code>
<code>hasNext()</code>	Checks if input is available.	<code>if(scanner.hasNext())</code>
<code>close()</code>	Closes the scanner to free resources.	<code>scanner.close();</code>

Reading Input from the User

Reading Different Data Types

```
import java.util.Scanner;

public class ScannerExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read integer
        System.out.print("Enter an integer: ");
        int number = scanner.nextInt();
        System.out.println("You entered: " + number);

        // Read double
        System.out.print("Enter a decimal number: ");
        double decimal = scanner.nextDouble();
        System.out.println("You entered: " + decimal);

        // Read string
        System.out.print("Enter a word: ");
        String word = scanner.next();
        System.out.println("You entered: " + word);
    }
}
```

```
// Read entire line
scanner.nextLine(); // Consume the leftover newline
System.out.print("Enter a sentence: ");
String sentence = scanner.nextLine();
System.out.println("You entered: " + sentence);

scanner.close();
}
}
```
